

Method Overloading

1. Write two methods with the same name but different number of parameters of same type and call the methods from main method

Answer:

```
class Demo {
    void add(int a) {
        System.out.println("One Parameter: " + a);
    }
    void add(int a, int b) {
        System.out.println("Two Parameters: " + (a + b));
    }
    public static void main(String[] args) {
        Demo obj = new Demo();
        obj.add(10);
        obj.add(10, 20);
    }
}
```

Output:

One Parameter: 10
Two Parameters: 30

2. Write two methods with the same name but different number of parameters of different data type and call the methods from main method

Answer:

```
class Demo {
    void display(int a) {
        System.out.println("Integer: " + a);
    }
    void display(int a, String name) {
        System.out.println(a + " " + name);
    }
    public static void main(String[] args) {
        Demo obj = new Demo();
        obj.display(100);
        obj.display(101, "Naveen");
    }
}
```

Output:

Integer: 100
101 Naveen

3. Write two methods with the same name and same number of parameters of same type and call from main method

Answer:

Not Possible in Java

```
class Demo {  
    void show(int a) { }  
  
    // ✗ Compile-time Error  
    void show(int b) { }  
}
```

Reason: Both methods have the **same method signature**.

4. Write two methods with the same name and same number of parameters of different type and call from main method

Answer:

```
class Demo {  
    void show(int a) {  
        System.out.println("Integer: " + a);  
    }  
    void show(String name) {  
        System.out.println("String: " + name);  
    }  
    public static void main(String[] args) {  
        Demo obj = new Demo();  
        obj.show(10);  
        obj.show("Naveen");  
    }  
}
```

Output:

Integer: 10
String: Naveen

5. Write two methods with the same name, number of parameters and data type but different return Type

Answer:

Not Possible in Java

```
class Demo {  
  
    int add(int a) {  
        return a;  
    }  
  
    // ✖ Compile-time Error  
    String add(int a) {  
        return "Hello";  
    }  
}
```

Reason: Java **does not allow method overloading based only on return type.**